Semantic Search and Change Detection in Aerial Imagery

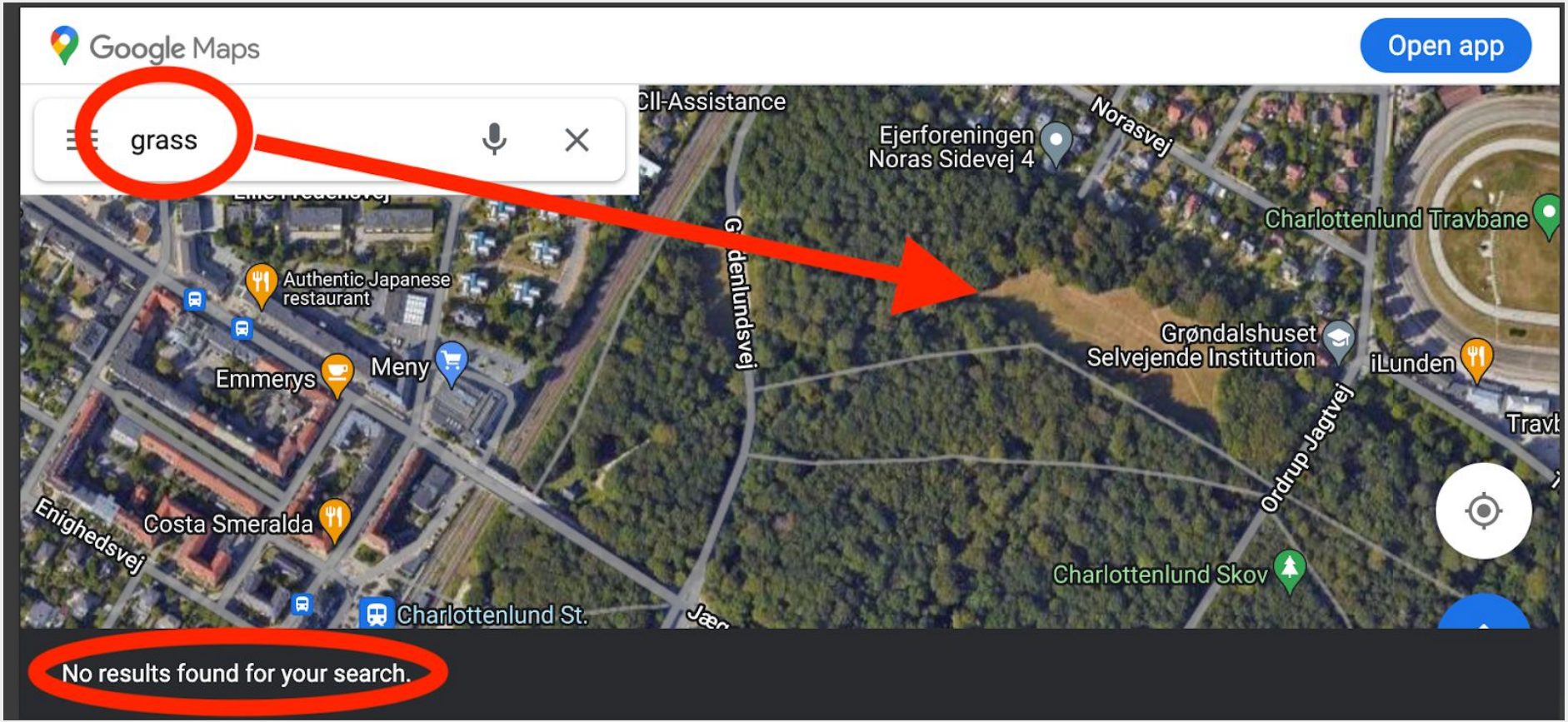
A practical look at vector embeddings

Matthew Hanson
Principal Applied AI Engineer

STRATUS Conference
May 17, Rochester, NY

1.

The Imagery Search Problem



Why can’t Google find grass?

Filters | Item Details

1 scenes selected



STAC API Item

Title:
S2C_T17TQH_20260512T161030_L2A

Datetime:
2026-05-12T16:11:26.790000Z

Platform:
sentinel-2c

Eo:cloud_cover:
8.79 %

1 of 1

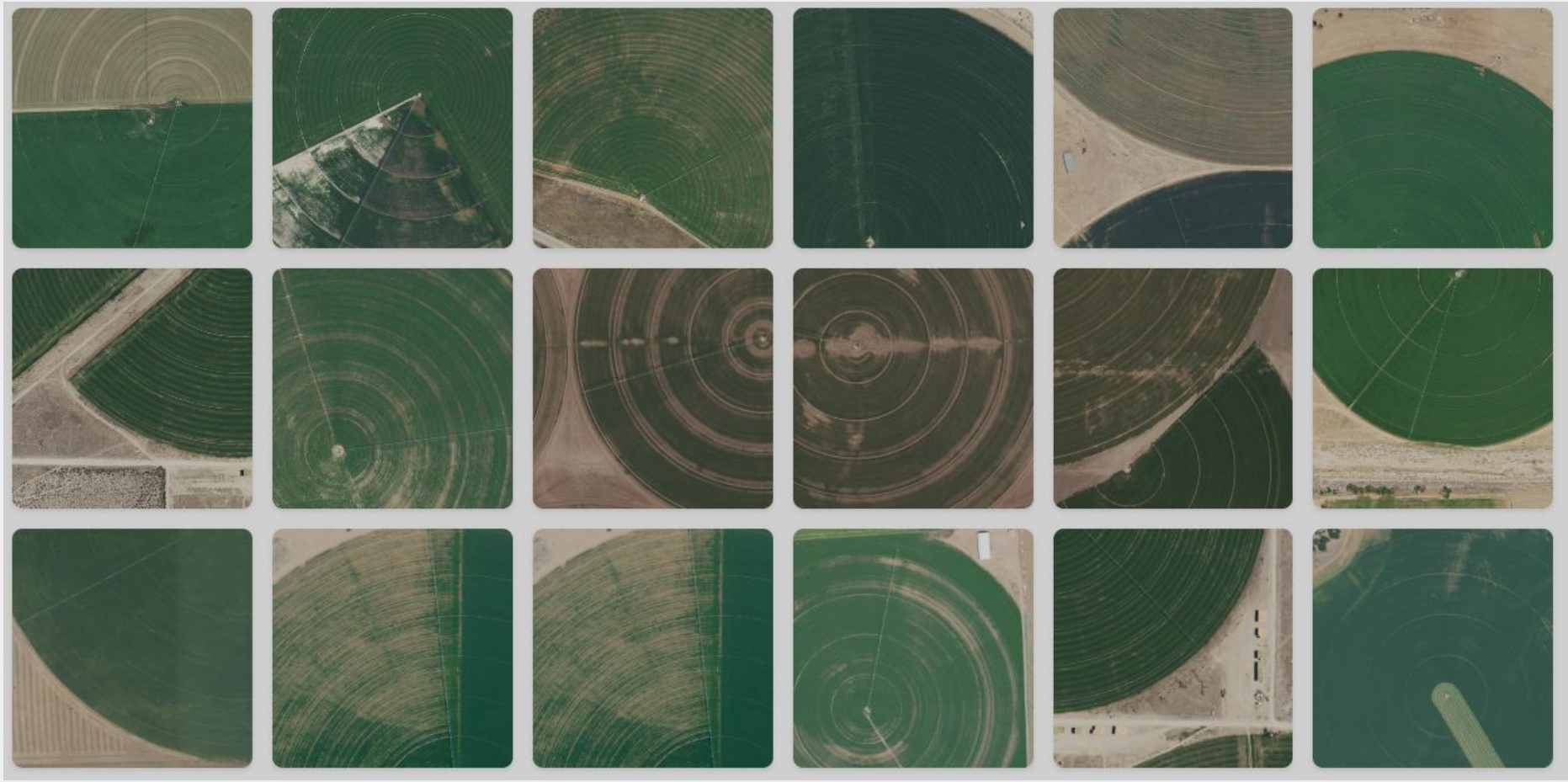
Showing 21 of 21 scenes

Select scenes

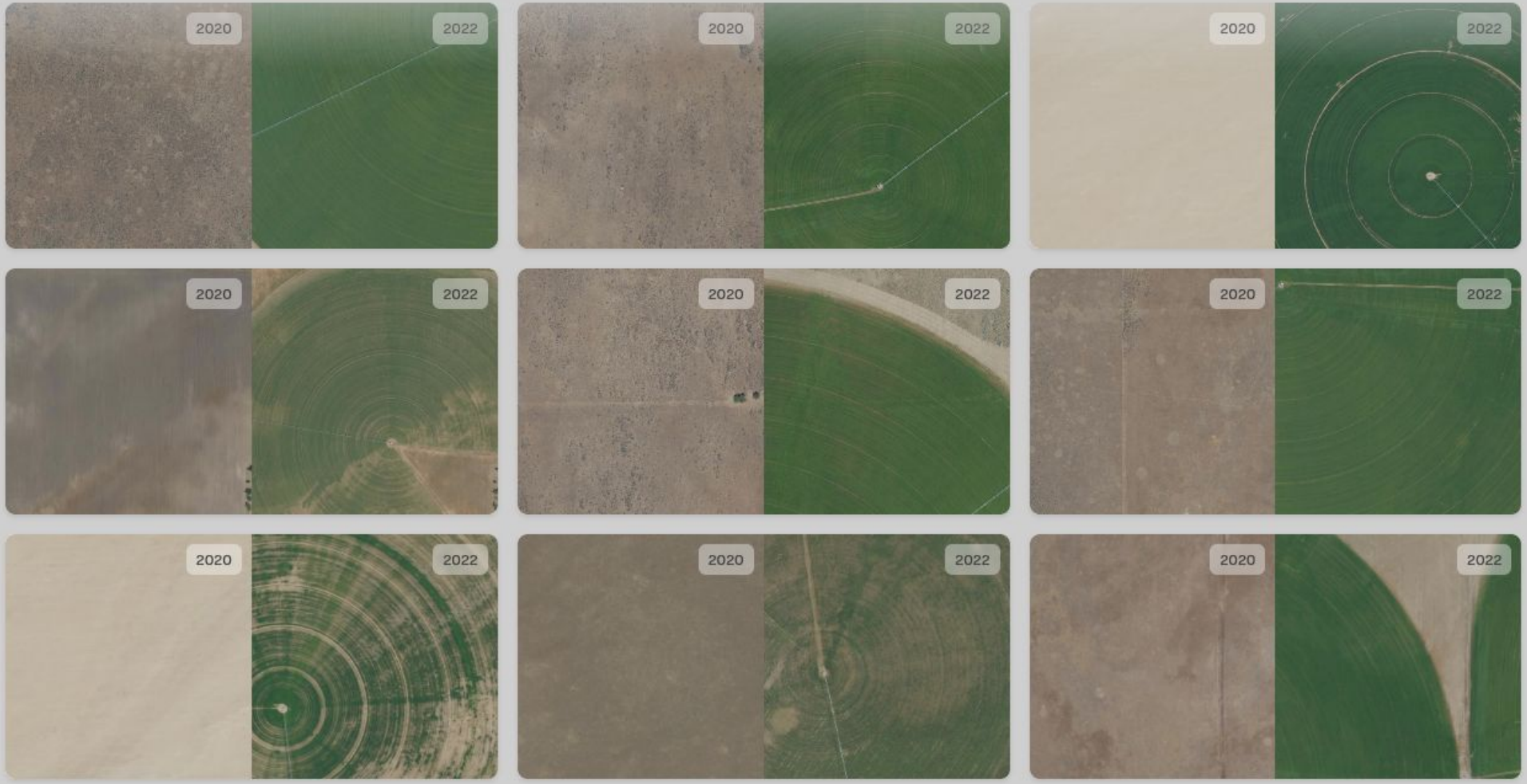
Available scene

Search area

Niagara-on-the-Lake
St. Catharines
Lockport
Niagara Falls
Pelham
North Tonawanda
Welland
Amherst
Port Colborne
Buffalo
Depew
West Seneca
East Aurora
Hamburg
Springville
Gowanda
Ellicottville
Salamanca
Newark
Seneca Falls
Geneva
Penn Yan
Watkins Glen
Horseheads



The challenge with aerial imagery is no longer indexing it.
It's finding the thing you actually care about inside it — and recognizing when it's changed.

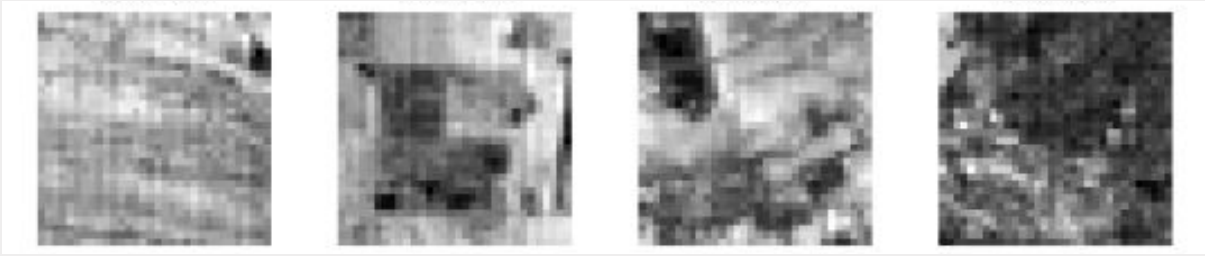

2020
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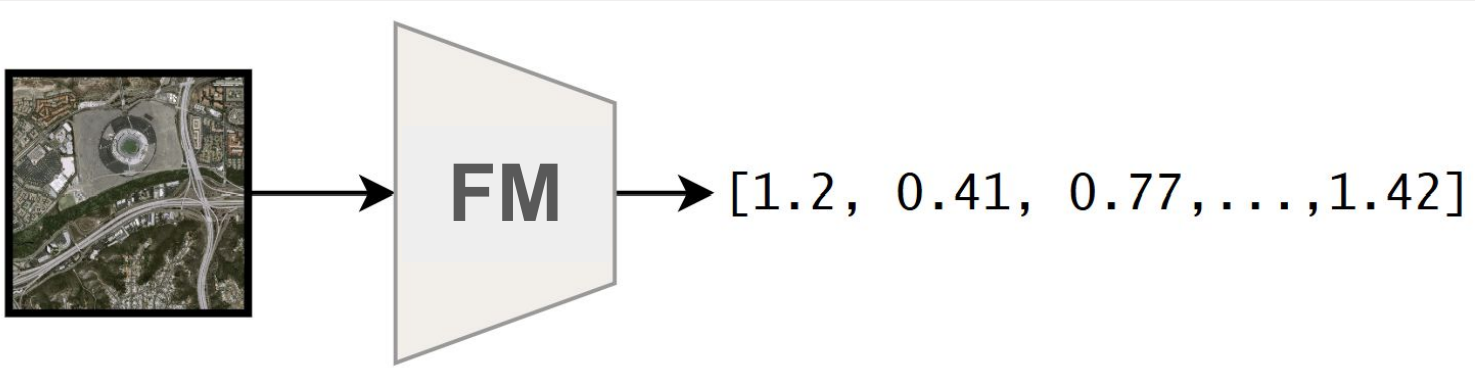
2.

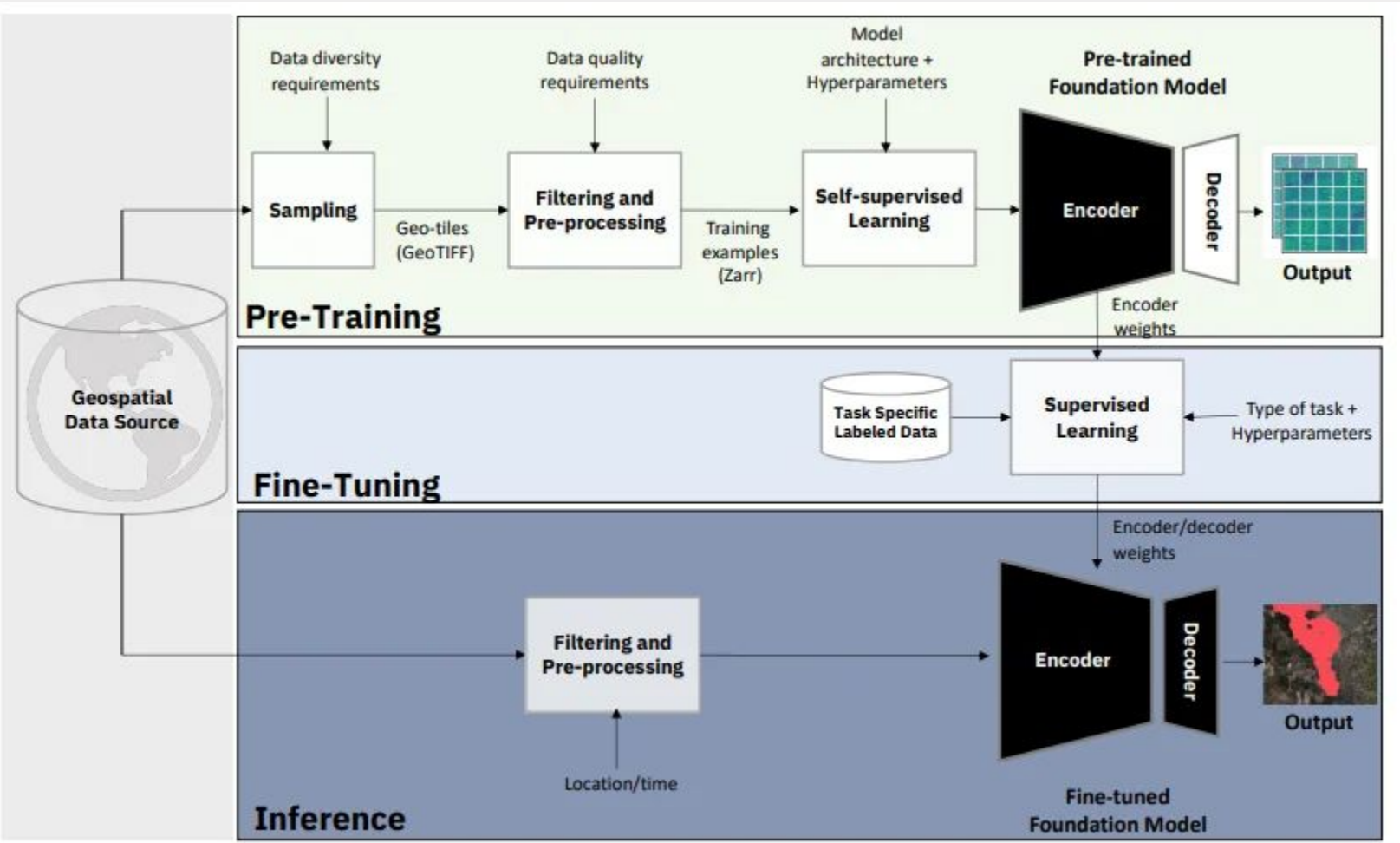


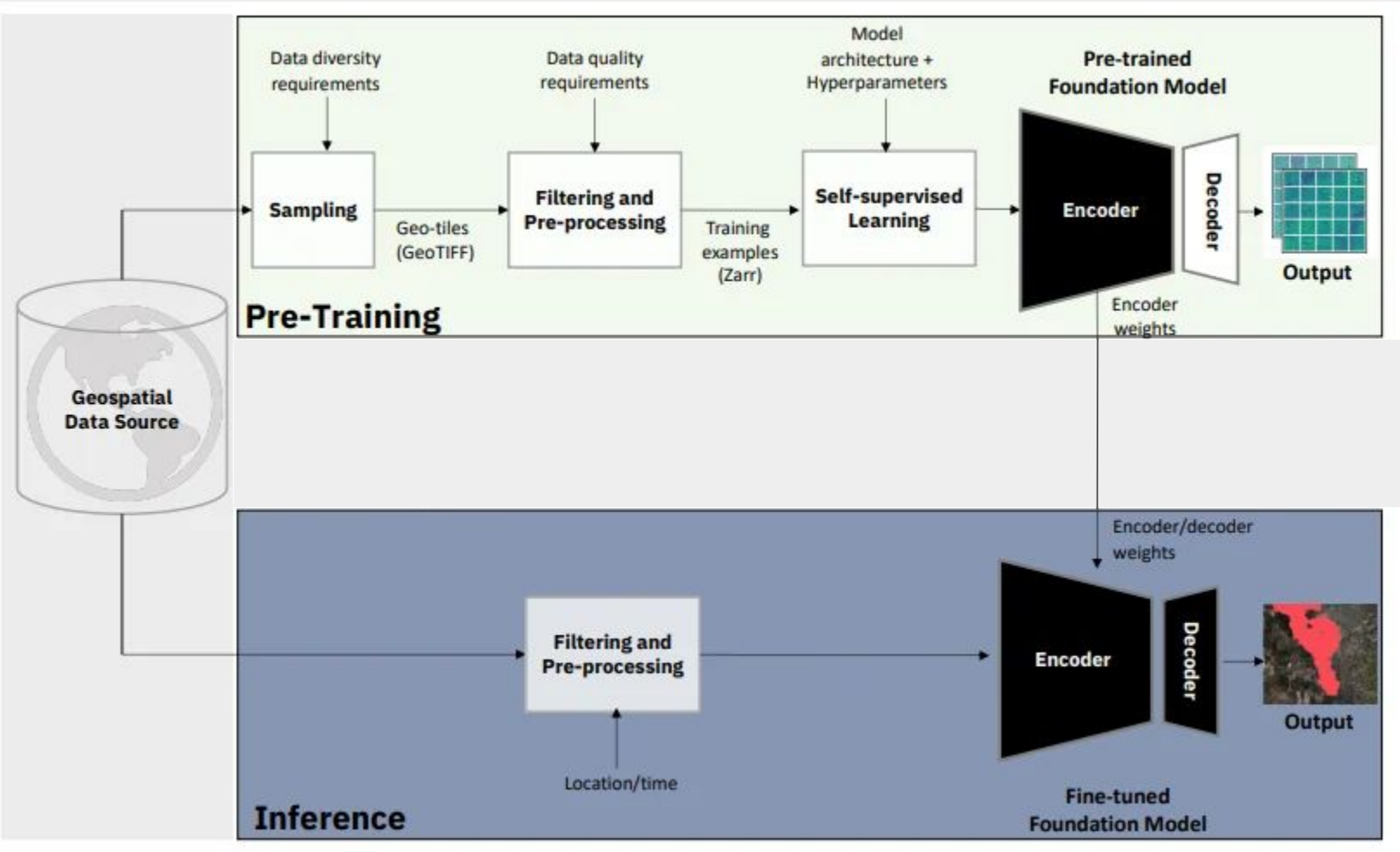
Foundation Models



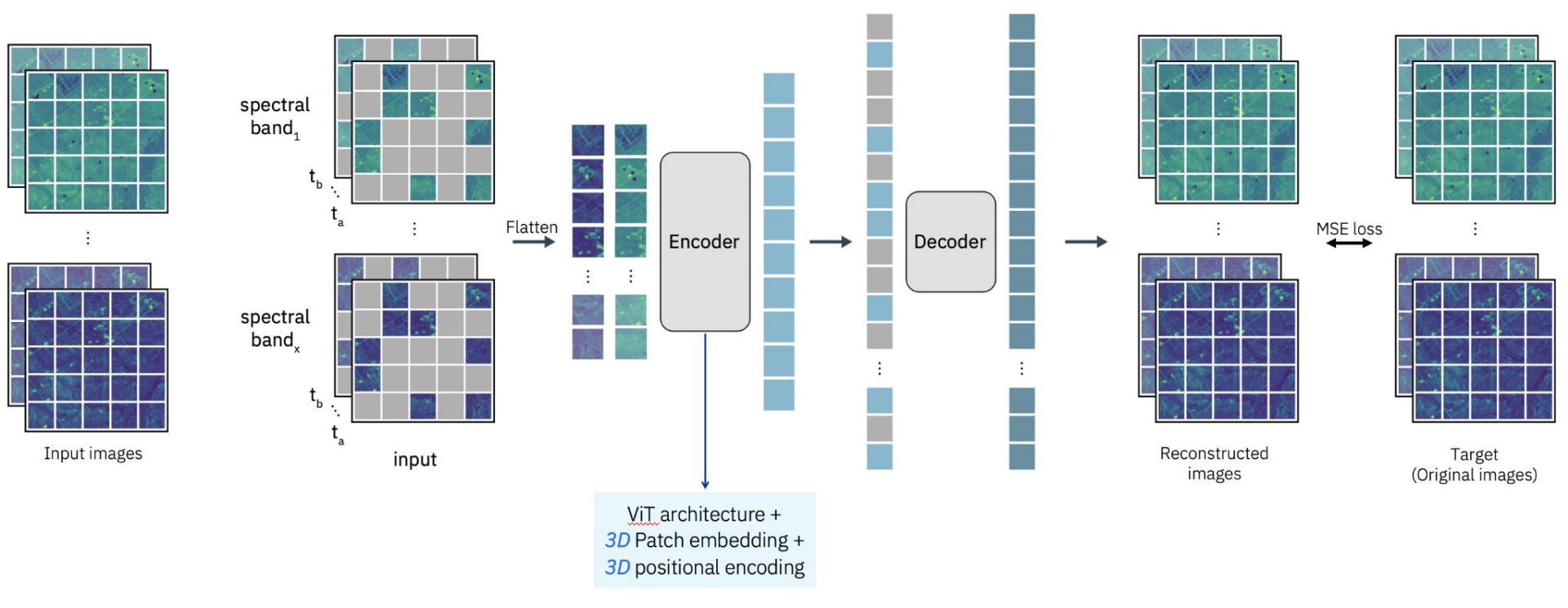



FM
[1.2, 0.41, 0.77,...,1.42]


Data diversity requirements
Data quality requirements
Model architecture + Hyperparameters
Pre-trained Foundation Model
Sampling
Geo-tiles (GeoTIFF)
Filtering and Pre-processing
Training examples (Zarr)
Self-supervised Learning
Encoder
Decoder
Output
Pre-Training
Encoder weights
Geospatial Data Source
Task Specific Labeled Data
Supervised Learning
Type of task + Hyperparameters
Fine-Tuning
Encoder/decoder weights
Filtering and Pre-processing
Encoder
Decoder
Output
Location/time
Fine-tuned Foundation Model
Inference


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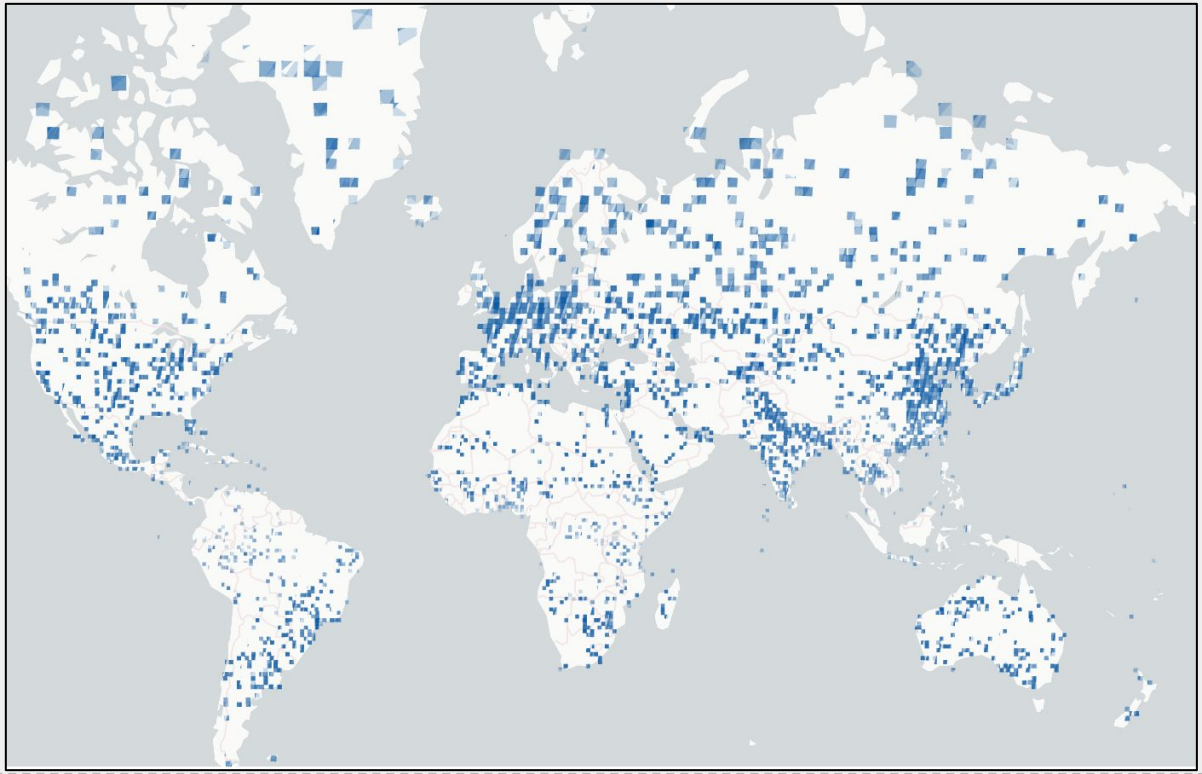
Prithvi (IBM and NASA's model)



CLAY, AI FOR EARTH

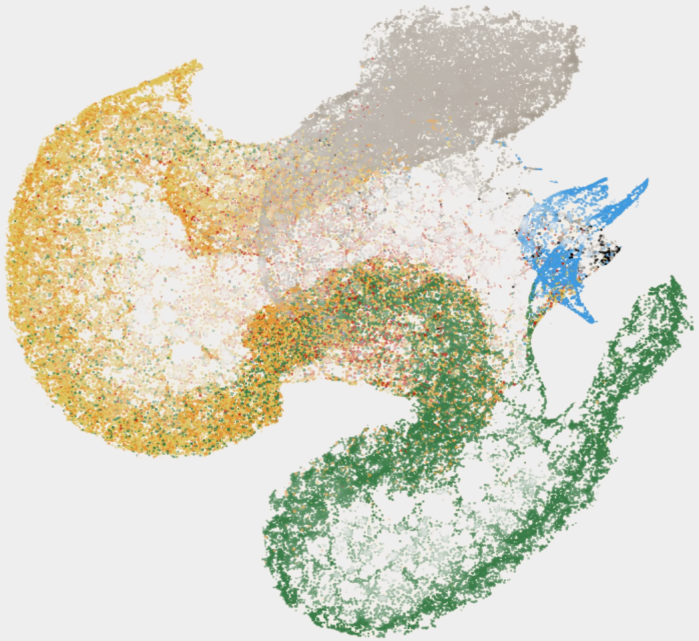
Clay Model v1

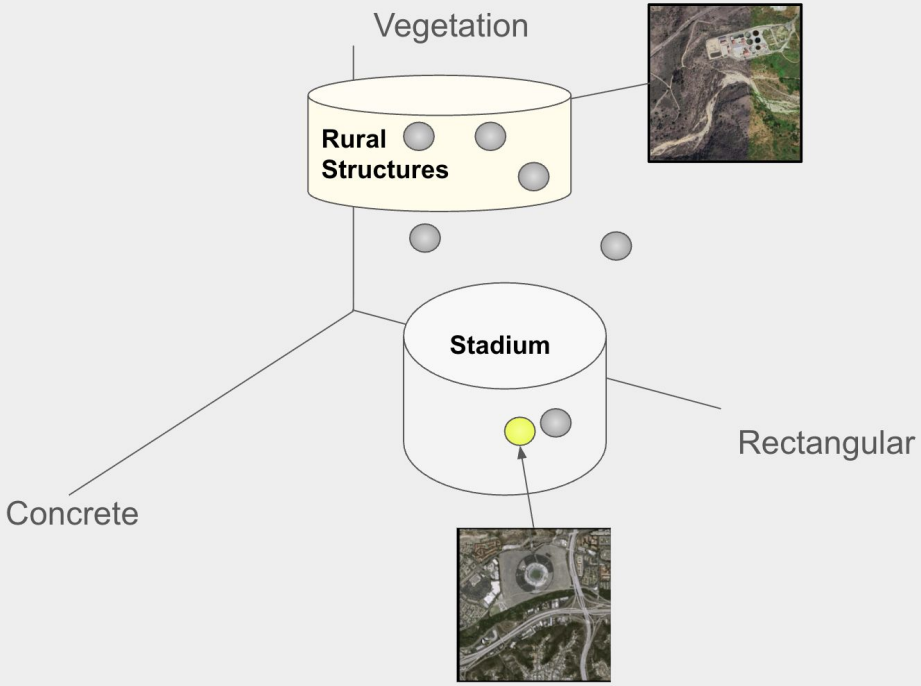
- ➔ Trained on 70 million satellite images, from 2018 to 2023
- ➔ Resolutions range from 30 meters to 30 centimeters
- ➔ 4000 GPU hours on H100 (p5.48xlarge) servers,
- ➔ 35 trillion pixels, 201 million parameters



3.

Vector Embeddings



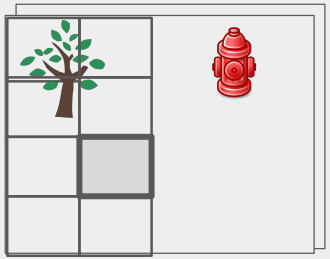


Embeddings translate pixels into conceptual “vectors” that can be clustered for quick discovery and retrieval.

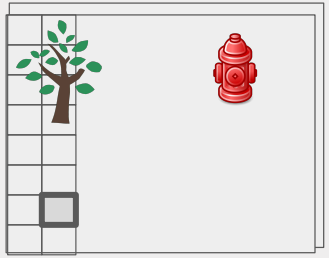
Chip Sizes

Variable Chip Size

Large Chip Size



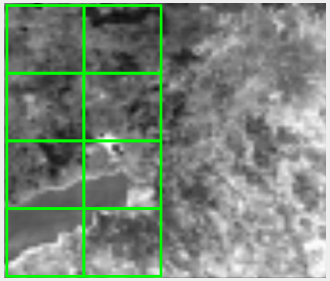
Smaller Chip Size

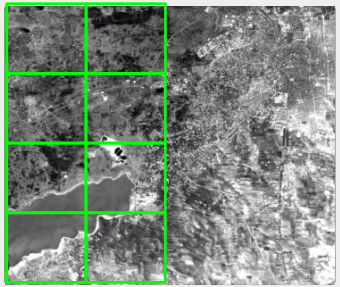


Any Spatial Resolution

Cm, m, km

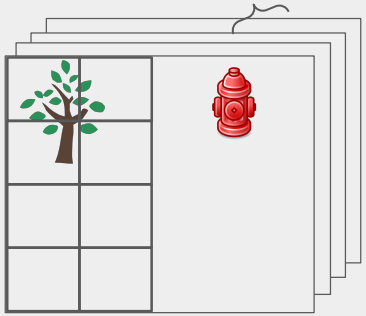




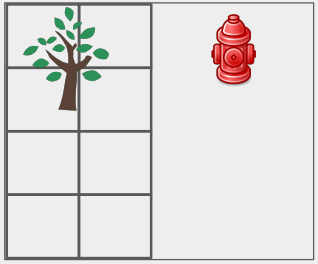


Wavelength Agnostic

Multi Band

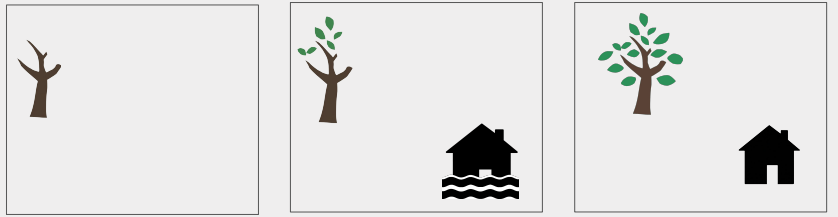


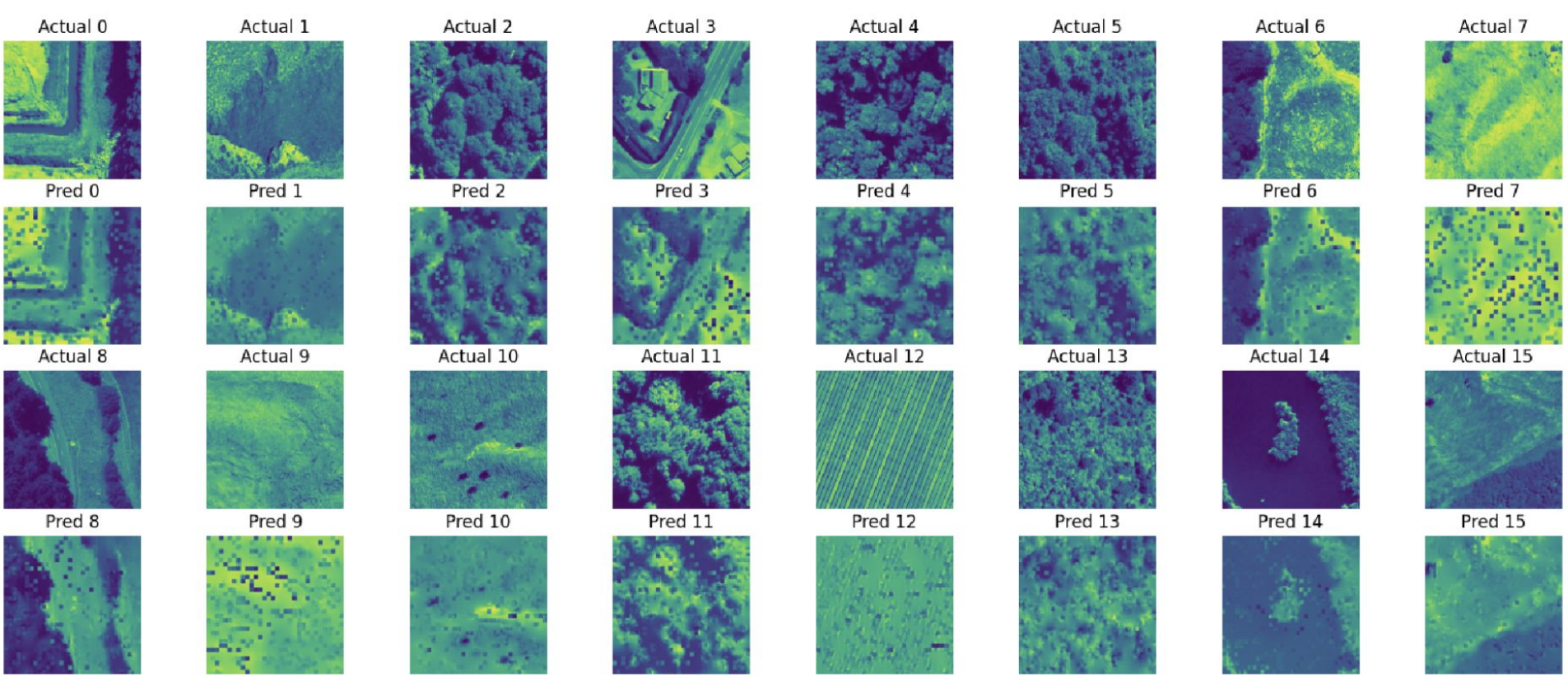
Single Band



Any Date/Acquisition

Capture changes at any temporal resolution





Reconstructions

Vector Similarity

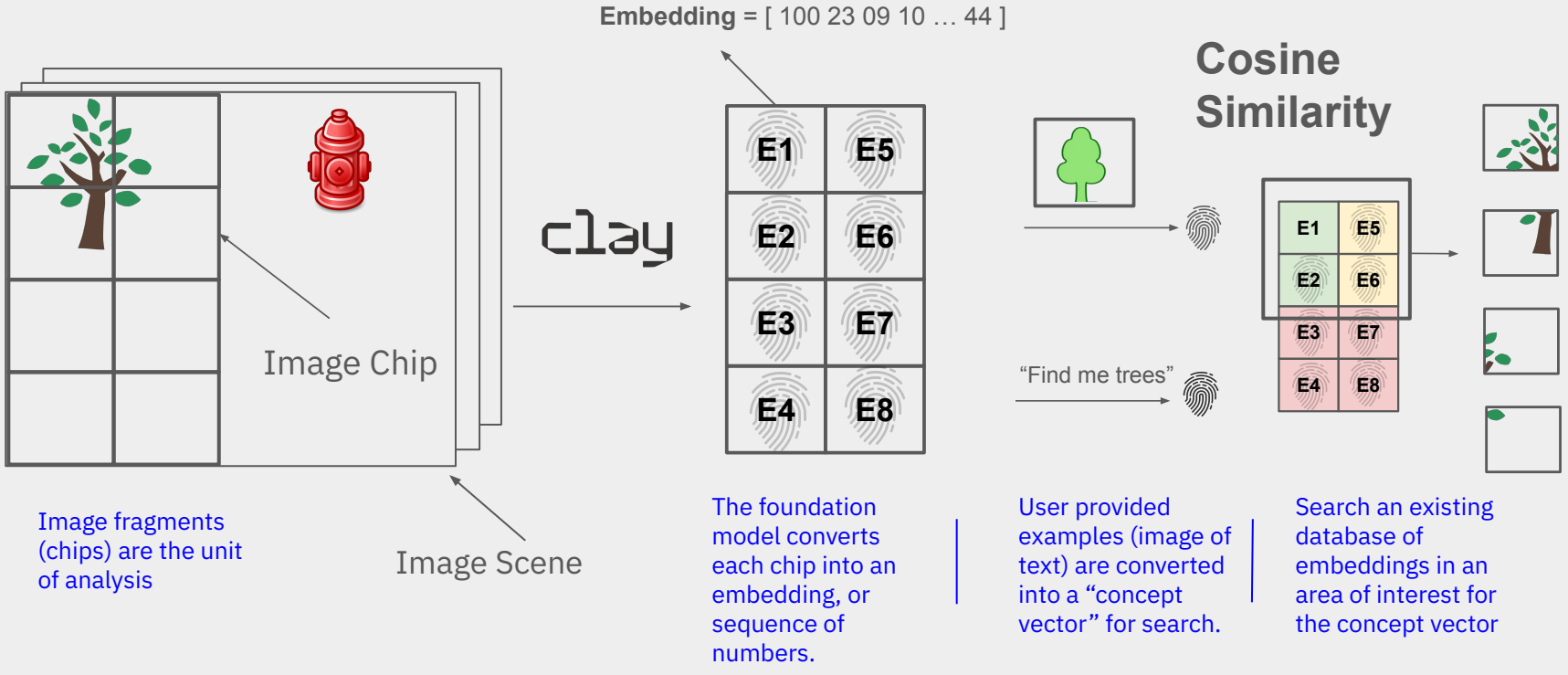
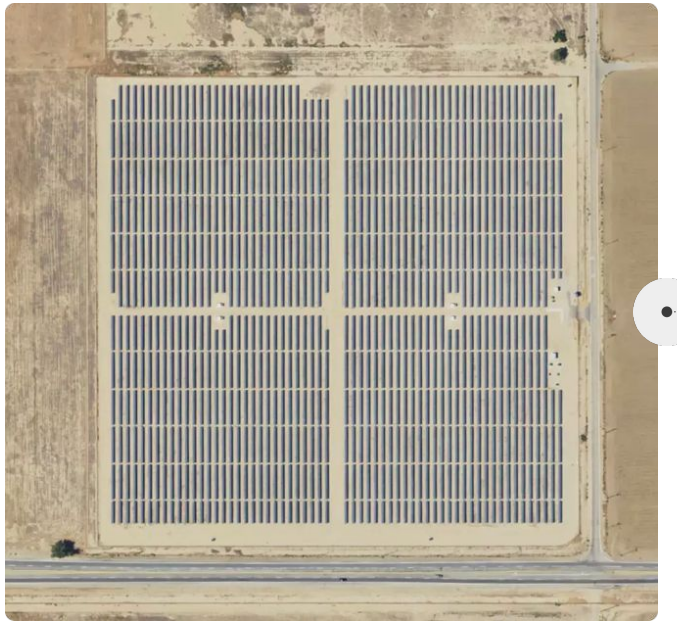


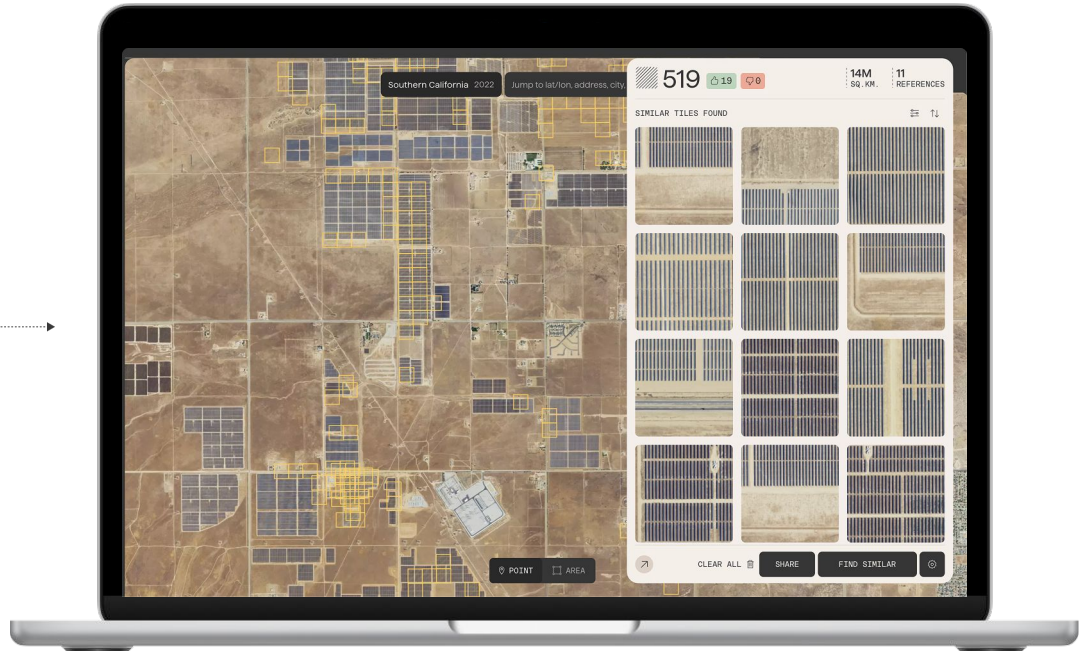
Image fragments (chips) are the unit of analysis

The foundation model converts each chip into an embedding, or sequence of numbers.

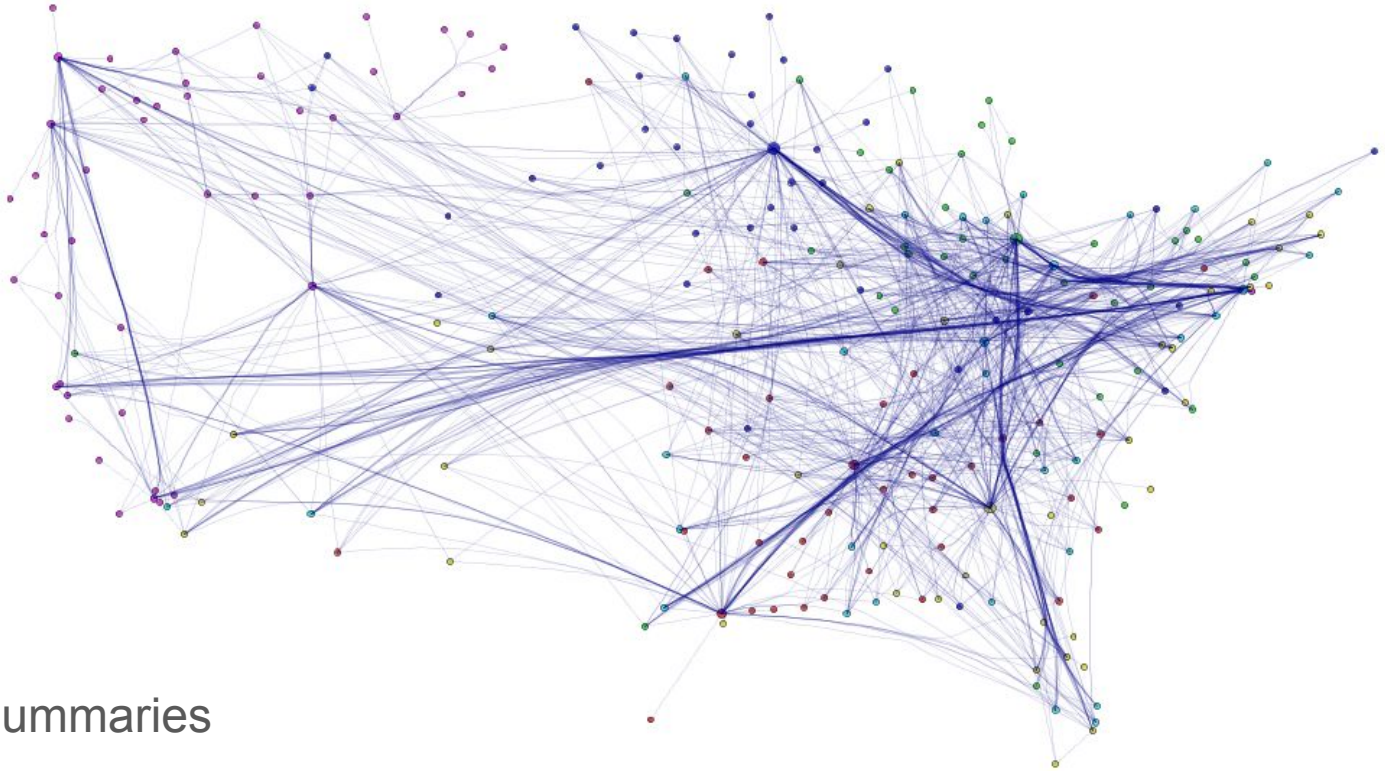
User provided examples (image of text) are converted into a "concept vector" for search.

Search an existing database of embeddings in an area of interest for the concept vector




Southern California 2022
Jump to lat/lon, address, city
519
19
0
14M
SQ. KM.
11
REFERENCES
SIMILAR TILES FOUND
POINT
AREA
CLEAR ALL
SHARE
FIND SIMILAR

Embeddings



They are summaries

They are imperfect summaries

They are imperfect but detailed summaries

They are imperfect but detailed mathematical summaries

```
The quick brown
fox...
```


observation
surface
detection
time
process
analysis
agricultural
missions
years
stac
moisture
earth
approach
david
performance
field
analysis-ready
pixel
global
satellite
analytics
talk
monitoring
water
sensors
point
imaging
ecosystem
reflectance
level
datasets
computing
discuss
sentinel-2
available
imagery
commercial
remote
sensing
applications
information
landsat
quality
data
products
land
spatial
product
fusion
classification
common
planetscope
tools
airbus
challenges
ard
earthdaily
constellation
community
development
approaches
due
learning
research
accuracy
assessment
system
joe
ready
phenology
improve
soil
provide
geometric
xarray
sources
chris
basemaps
agriculture
peatlands
arcgis
temporal
new
systems
crop
satellites
processing
sen2like
observations

Question shape	Right substrate
"Find me places that look like this"	Embeddings
"Find sites where X is happening"	Embeddings
"Has this region changed semantically?"	Embeddings
"Count instances of a phenomenon"	Embeddings + segmentation
"Search across a large archive in seconds"	Embeddings (only feasible at scale)
"Measure surface reflectance / temperature / biomass"	ARD
"Has this region changed quantitatively (N% biomass loss)?"	ARD
"Authoritative measurement for regulatory work"	ARD
"Calibrated long-baseline continuity record"	ARD

4.

Open-Source models for UAV Imagery

Model	Trained on	UAV applicability
Clay v1.5	Sentinel-2, Sentinel-1, Landsat — multi-sensor satellite	Transfer to UAV is empirical and untested at scale; closer-to-domain than purely Sentinel models
AlphaEarth Foundations (Google DeepMind)	Multi-sensor satellite cube; pixel-level embeddings	Annual global maps 2018–2024; not UAV-specific; pixel-level granularity conceptually well-aligned with UAV detail
Prithvi (IBM/NASA)	HLS (Harmonized Landsat-Sentinel)	Satellite-trained; transfer to UAV unlikely strong without fine-tuning
SatlasPretrain (AllenAI)	**Includes aerial imagery alongside Sentinel-1/2 and Landsat**	The published EO model with the most plausible UAV transfer story today
RemoteCLIP	Remote-sensing image-caption corpus	Text-aligned (Pillar 3); aerial imagery in training is partial
Copernicus-FM	Copernicus Sentinel missions, multimodal	Satellite-only; not UAV-applicable directly
General vision FMs (DINOv2, CLIP, SigLIP)	Internet-scale natural images	Often transfer surprisingly well to UAV — UAV imagery sits closer to natural-image distribution than satellite does

Foundation Model Takeaways

- There is no UAV foundation model with broad public adoption today
- SatlasPretrain is the most promising satellite-foundation candidate
- General vision foundation models are a serious option for UAV
- Domain-specific fine-tuning is the gap-closer

Model Evaluation for UAV Imagery

1. **Spectral match** — RGB vs. multi-spectral
2. **Spatial scale** — training resolution vs. UAV resolution
3. **Geometry** — nadir vs. oblique training data
4. **Domain familiarity** — did it see imagery with the kinds of features UAV captures?
5. **Text alignment support**

5.

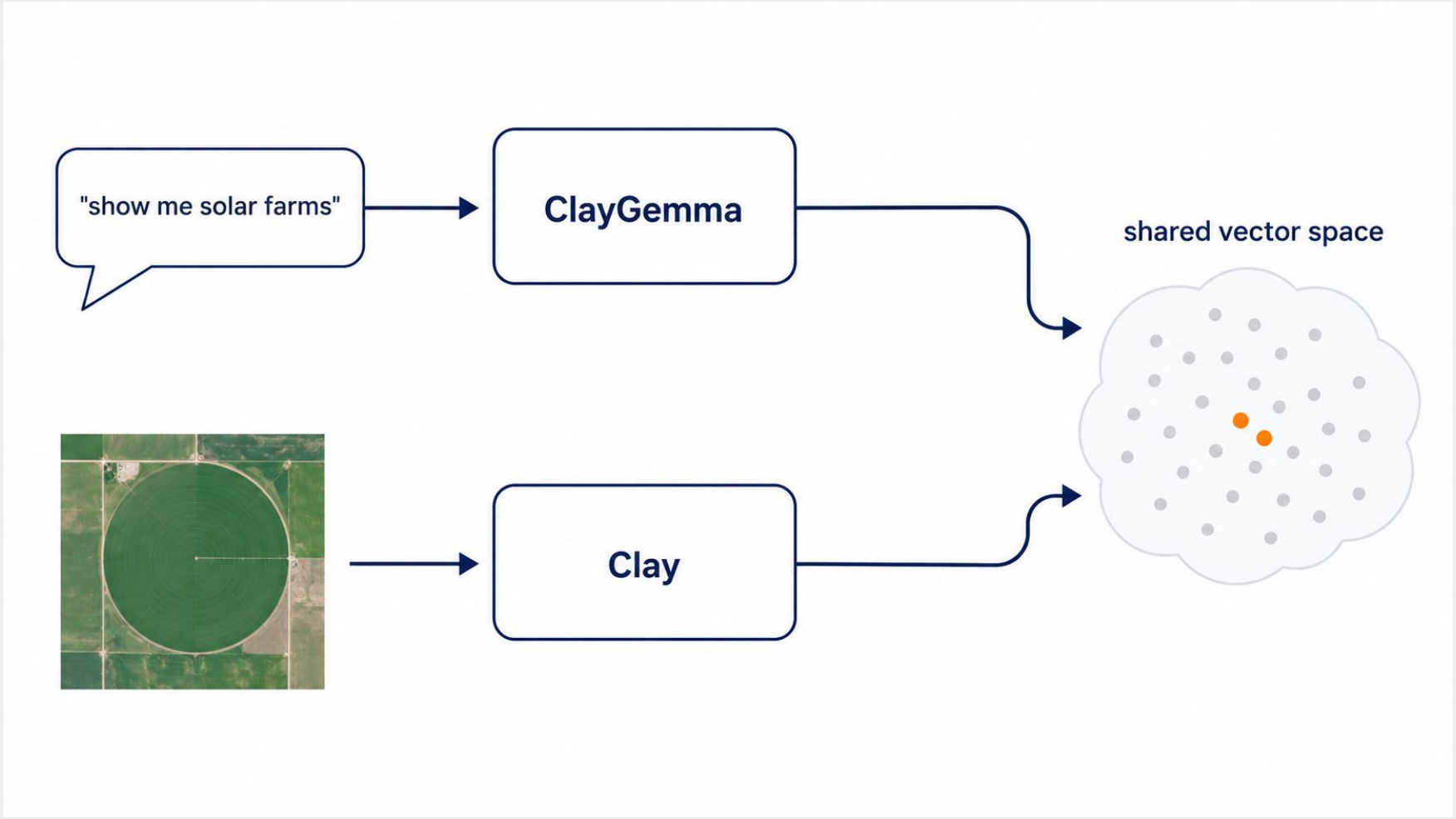


Text-Imagery vector alignment

- An image encoder turns imagery into a vector.
- A *text encoder* turns text into a vector
    - in the *same* space, if the two encoders were trained to align.

Two paths to alignment:

- Joint training (CLIP-style)
    - vision and language encoders trained simultaneously on image-caption pairs (RemoteCLIP)
- Bridging models
    - train text encoder to align to existing vision model embedding space (LGND's ClayGemma, OpenTextBridge)



text alignment turns a static image index into a dynamic query

“Show me planes”



33.9475° N, 118.4058° W

Los Angeles, California 90045, United States
Clay 1.0 NAIP 2022 256px

“I only want planes at the gate”



33.9405° N, 118.4090° W

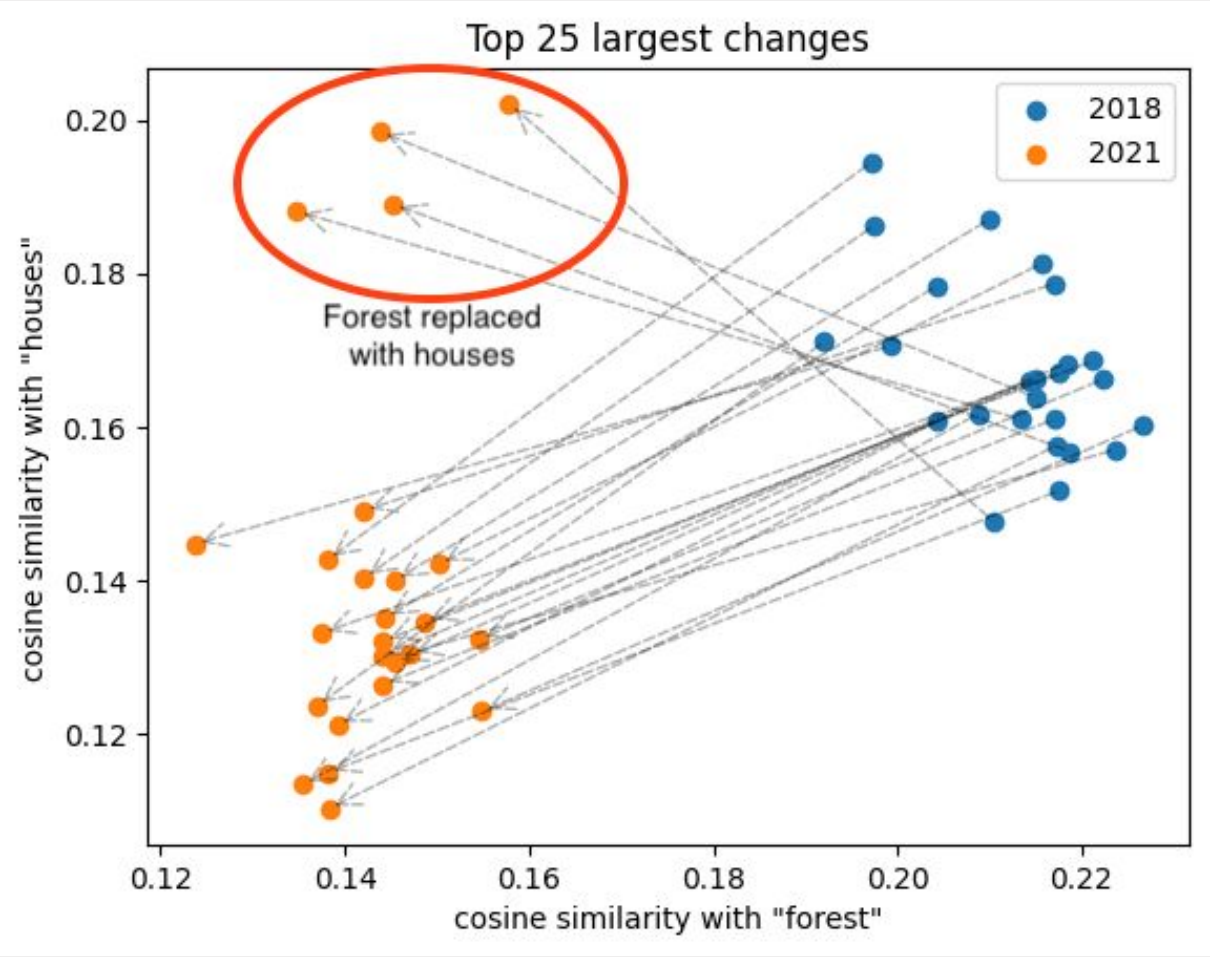
Los Angeles, California 90045, United States
Clay 1.0 NAIP 2022 256px

6.

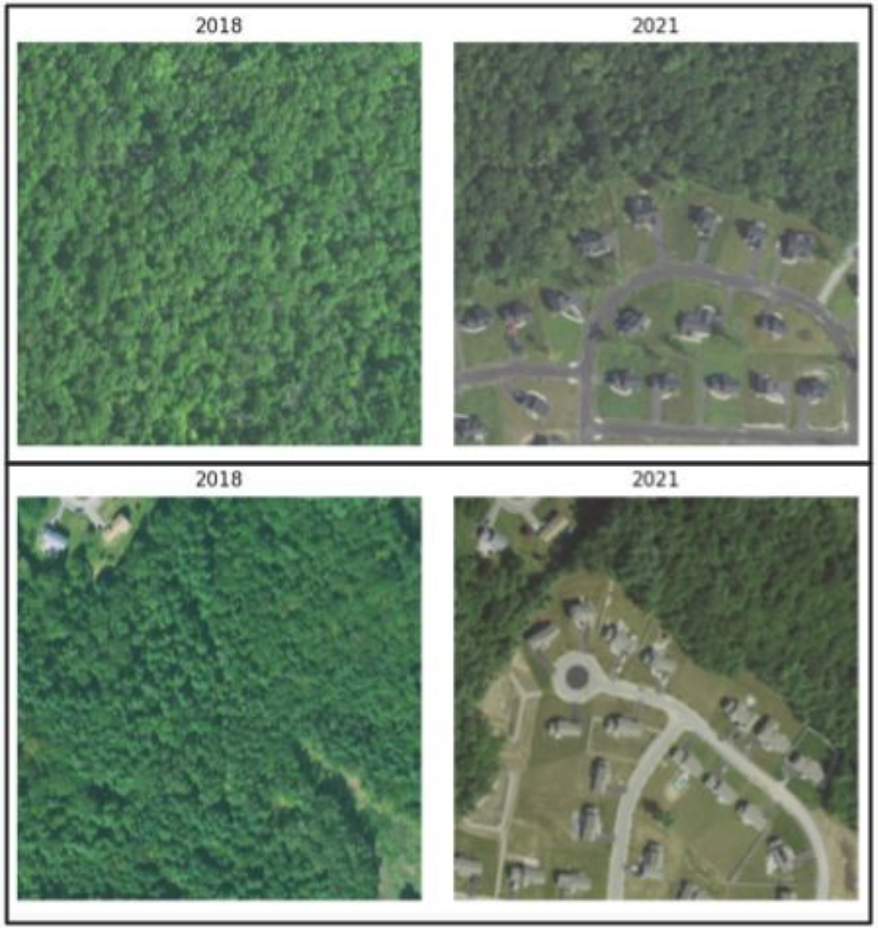
Change detection in vector space

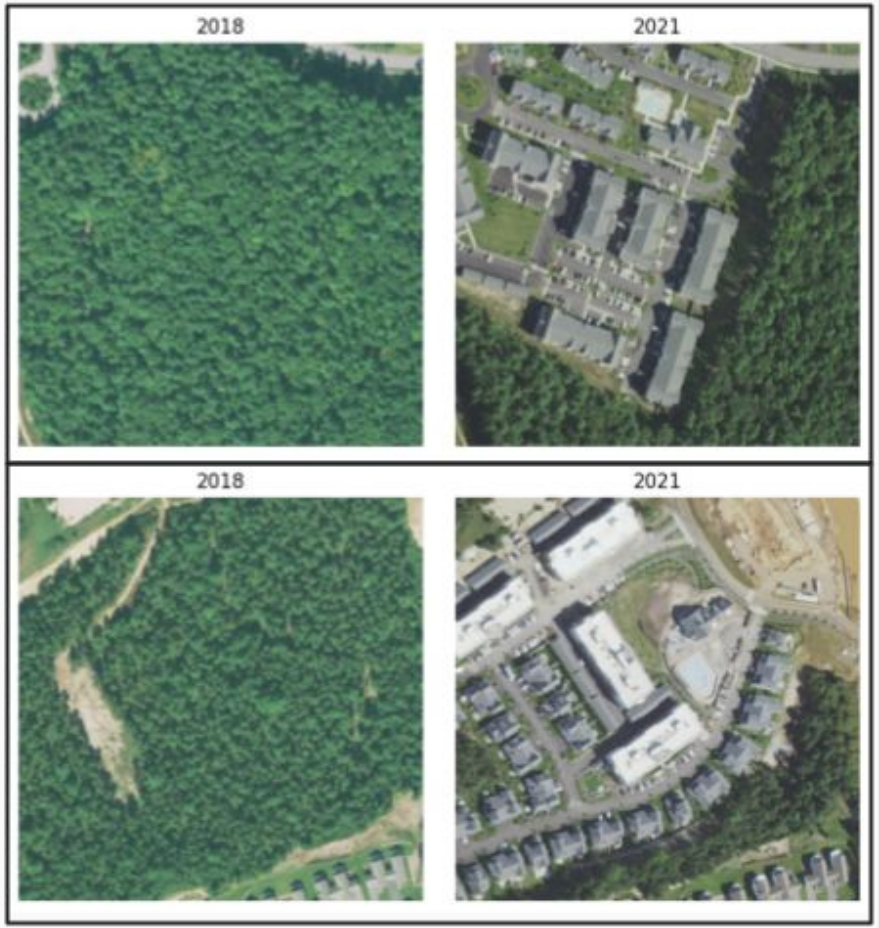
“Show me locations that transition from farmland to residential”





https://element84.com/machine-learning/finding-changes-on-the-earth-with-natural-language/





https://element84.com/machine-learning/finding-changes-on-the-earth-with-natural-language/

“Show me new housing developments in California”



CHANGE DESCRIPTION

Summary: The dominant regional change is widespread and rapid residential urban development transforming previously undeveloped land into new housing communities.

Dominant patterns:

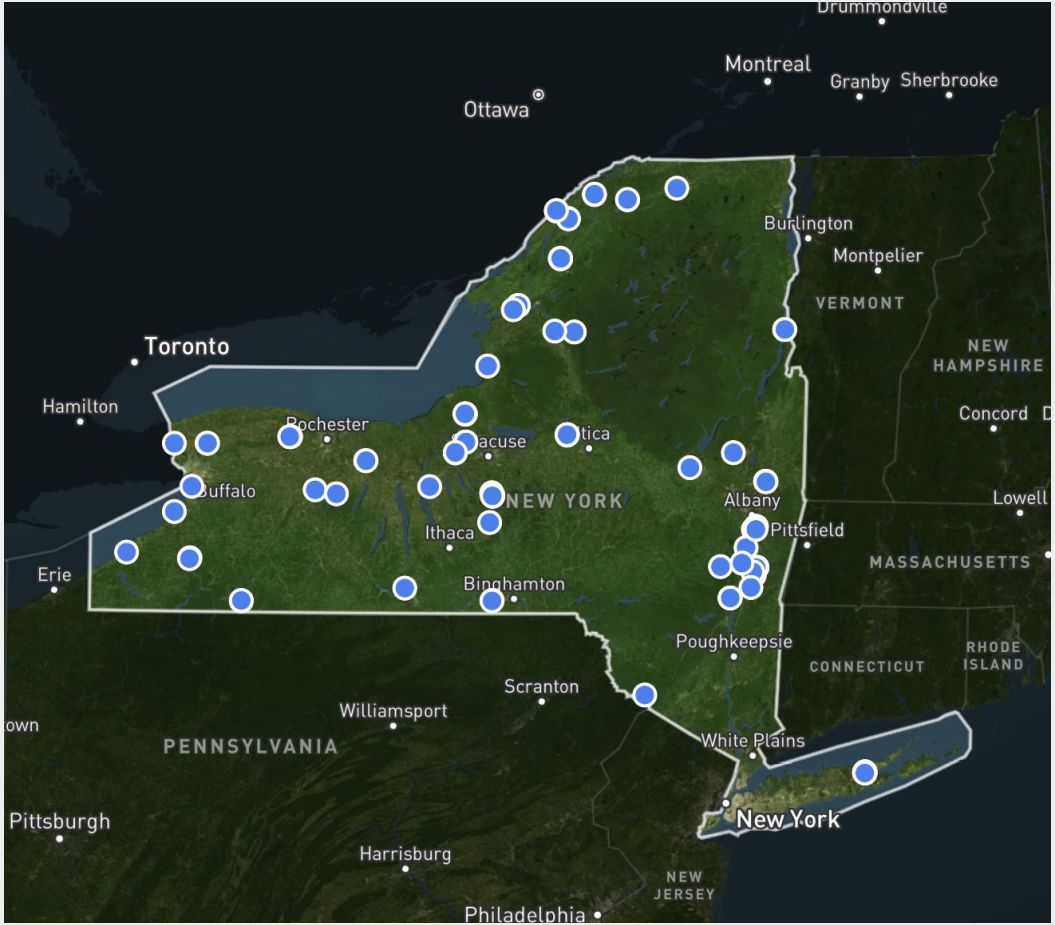
- **Urban growth** (throughout): Extensive residential housing construction and associated infrastructure development are evident across all sites.
- **Land transformation** (throughout): Bare earth and initial grading have been replaced by dense networks of paved roads, houses, and communal green spaces.

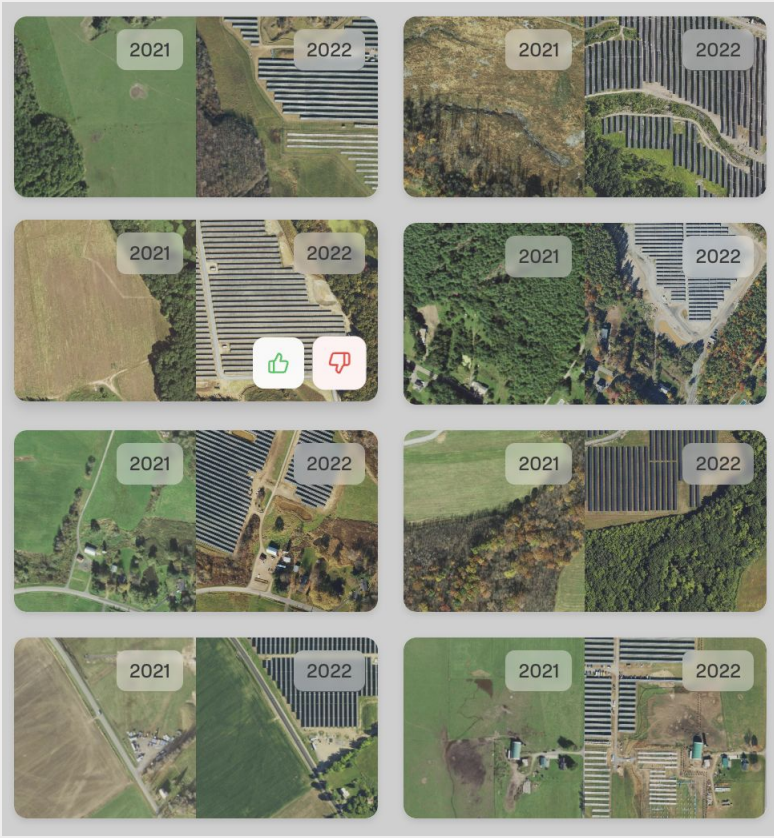
Most-changed pairs:

- **Pair 1** (major): Transformation from bare earthworks and graded plots to a dense residential neighborhood with completed homes and paved roads.
- **Pair 2** (major): Extensive residential development with numerous houses constructed on previously graded land.
- **Pair 3** (major): Development from graded plots to a mature residential area featuring completed houses, paved roads, and a large central community park.

- The same vector index that powers search powers change detection
- Ranks **semantically meaningful** change, not every-pixel difference.
- Shadows, seasonal vegetation, atmospheric variation, registration noise don't dominate the signal the way they do in pixel space.

find new solar arrays in New York State between 2020 and 2024





Takeaways

- Embeddings make imagery semantically searchable
- The OS model landscape for UAV is workable but incomplete
- Text alignment is the interface
- Change detection in vector space ranks meaning, not pixels

Explore

LGND Discover - https://discover.lgnd.ai/

Chip Match Game - https://chipmatch.apps.lgnd.hosted.unionai.cloud/

E84 Queryable Earth - https://element84.com/queryable-earth/

Earth Genome Earth Index - https://www.earthgenome.org/earth-index